

**AMMONIUM HYDROXIDE (10%-35% solution)****ICSC: 0215 (May 2018)**

Aqua ammonia

Ammonium hydrate

Ammonia aqueous solution (10-35%)

**CAS #: 1336-21-6****UN #: 2672****EC Number: 215-647-6**

|                             | ACUTE HAZARDS               | PREVENTION | FIRE FIGHTING                                                                                                                             |
|-----------------------------|-----------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>FIRE &amp; EXPLOSION</b> | Not combustible. See Notes. |            | In case of fire in the surroundings, use appropriate extinguishing media. In case of fire: keep drums, etc., cool by spraying with water. |

**STRICT HYGIENE! IN ALL CASES CONSULT A DOCTOR!**

|                   | SYMPTOMS                                                                        | PREVENTION                                                                   | FIRST AID                                                                                                                                |
|-------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Cough. Sore throat. Burning sensation. Laboured breathing. Shortness of breath. | Use ventilation, local exhaust or breathing protection.                      | Fresh air, rest. Artificial respiration may be needed. Refer for medical attention.                                                      |
| <b>Skin</b>       | Redness. Pain. Blisters. Serious skin burns.                                    | Protective gloves. Protective clothing.                                      | Remove contaminated clothes. Rinse skin with plenty of water or shower for at least 15 minutes. Refer immediately for medical attention. |
| <b>Eyes</b>       | Redness. Pain. Blurred vision. Severe burns.                                    | Wear face shield or eye protection in combination with breathing protection. | Rinse with plenty of water for several minutes (remove contact lenses if easily possible). Refer immediately for medical attention.      |
| <b>Ingestion</b>  | Sore throat. Vomiting. Abdominal pain.                                          | Do not eat, drink, or smoke during work.                                     | Rinse mouth. Do NOT induce vomiting. Give one or two glasses of water to drink. Refer for medical attention.                             |

| SPILLAGE DISPOSAL                                                                                                                                                                                                                                                                                                                   | CLASSIFICATION & LABELLING                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evacuate danger area! Consult an expert! Personal protection: complete protective clothing including self-contained breathing apparatus. Ventilation. Do NOT let this chemical enter the environment. Cautiously neutralize spilled liquid with dilute acid such as dilute sulfuric acid. Wash away remainder with plenty of water. | <p>According to UN GHS Criteria</p> <div style="text-align: center;">  </div> <p><b>DANGER</b></p> <p>May be corrosive to metals<br/>Harmful if swallowed<br/>Causes severe skin burns and eye damage<br/>May cause respiratory irritation<br/>Very toxic to aquatic life</p> <p><b>Transportation</b><br/><b>UN Classification</b><br/>UN Hazard Class: 8; UN Pack Group: III</p> |
| STORAGE                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Store in an area without drain or sewer access. Separated from food and feedstuffs. See Chemical Dangers. Cool. Well closed. Keep in a well-ventilated room. Provision to contain effluent from fire extinguishing. See Notes.                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| PACKAGING                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Unbreakable packaging.<br>Put breakable packaging into closed unbreakable container.<br>Do not transport with food and feedstuffs.                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.  
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**AMMONIUM HYDROXIDE (10%-35% solution)****ICSC: 0215****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

VERY VOLATILE COLOURLESS AMMONIA SOLUTION IN WATER WITH PUNGENT ODOUR.

**Physical dangers****Chemical dangers**

Reacts with many heavy metals and heavy metal salts. This produces explosive compounds. Attacks many metals. This produces flammable/explosive gas (hydrogen - see ICSC 0001). It reacts violently with acids. This generates fire and explosion hazard. Decomposes on heating. This produces toxic and corrosive fumes including ammonia and nitrogen oxides. This generates toxic hazard.

Formula:  $\text{NH}_4\text{OH}$

Molecular mass: 35.1

Boiling point: 38°C (25%)

Melting point: -58°C (25%)

Relative density (water = 1): 0.9 (25%)

Solubility in water: miscible

Vapour pressure, kPa at 20°C: 48 (25%)

Relative vapour density (air = 1): 0.6

See Notes.

**EXPOSURE & HEALTH EFFECTS****Routes of exposure**

The substance can be absorbed into the body by inhalation of its aerosol or vapour and by ingestion.

**Effects of short-term exposure**

The substance is corrosive to the eyes, skin and respiratory tract. Corrosive on ingestion. Inhalation of high concentrations of the vapour may cause laryngeal oedema, inflammation of the respiratory tract and pneumonia. Exposure could cause asphyxiation due to swelling in the throat. The effects may be delayed.

**Inhalation risk**

A harmful contamination of the air can be reached very quickly on evaporation of this substance at 20°C.

**Effects of long-term or repeated exposure**

Repeated or prolonged inhalation may cause effects on the lungs.

**OCCUPATIONAL EXPOSURE LIMITS**

TLV: 18 mg/m<sup>3</sup>, 25 ppm, as TWA; 27 mg/m<sup>3</sup> as STEL.

MAK: 14 mg/m<sup>3</sup>, 20 ppm; peak limitation category: I(2); pregnancy risk group: C

**ENVIRONMENT**

The substance is very toxic to aquatic organisms.

**NOTES**

Be aware that ammonia gas can evolve from ammonia solution.

Ammonia vapour is flammable and explosive under certain conditions.

See ICSC 0414.

Do NOT completely fill bottles with the substance; strong solutions may develop pressure.

Release caps with care.

Melting points range from -3°C (4%) to -69°C (28%). Another boiling point is 25°C (32%).

Other UN numbers are: UN 2073 Ammonia solution, relative density of less than 0.880 at 15°C in water, with more than 35% but no more than 50% ammonia; UN 3318 Ammonia solution, relative density of less than 0.880 at 15°C in water, with more than 50% ammonia.

Depending on the degree of exposure, periodic medical examination is suggested.

**ADDITIONAL INFORMATION****EC Classification**

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